Chenrui Ma

+1-949-792-8518 | zzz8fa@virginia.edu | Homepage | Scholar | GitHub
Charlottesville, Virginia, USA

Research Profile

First-year Computer Science Ph.D. student at the University of Virginia specializing in **efficient and reliable generative AI**, particularly diffusion and flow matching. My current research focuses on **reinforcement learning for generative-model post-training**. My emerging research interests include **continuous and discrete diffusion language models**. Seeking **Summer 2027 research internships**.

First-author papers: NeurIPS 2026 (2), CVPR 2026, AAAI 2026; co-first author: ACL 2026.

Education

University of Virginia *August 2026 – June 2030 (expected)*
Ph.D. in Computer Science; Advisor: Prof. Ferdinando Fioretto

University of California, Irvine *September 2025 – June 2026*
M.S. in Computer Science; GPA: 4.00/4.00; Advisor: Prof. Yanning Shen

University of California, Irvine *September 2024 – June 2025*
Exchange Student, UCI 3+2 Engineering Program in Computer Science; GPA: 3.92/4.00

Central South University *September 2021 – June 2025*
B.S. in Computer Science

Selected Research Projects

Efficient Generative Modeling

- Developed *Drift Flow Matching* [1], connecting direct generation with iterative flow models through drift-supervised transport, enabling **one-step generation and multi-step refinement** within one framework.
- Developed *Straight Variational Flow Matching* [3] for **efficient, high-quality sampling**, using a global latent code to disambiguate intersecting paths and a straightness objective to reduce trajectory curvature.
- Enabled **coarse-to-fine pixel-space generation** in a single end-to-end model by designing conditional dependent coupling [2] with shared parameters across resolutions.

Uncertainty-Aware Generative Modeling [7]

- Developed *Posterior Flow Matching*, a general generative-modeling method providing **generation and endpoint-uncertainty diagnostics in the same model evaluation**.
- Jointly learned conditional velocity means and variances with a single Gaussian likelihood objective; derived mean-driven ODE sampling and posterior-driven SDE sampling with uncertainty-adaptive noise.
- Demonstrated improved generation quality across images, 3D point clouds, molecular conformations, and robotic actions, with endpoint diagnostics throughout generation.

Generative Model Post-Training *Ongoing*

- Developed a **unified regression framework** for diffusion and flow model post-training, expressing existing methods through a common formulation that separates reward feedback from reference-model retention.
- **Derived reward-weighted flow matching from first principles** of KL-regularized reward maximization, connecting preference alignment to simple supervised regression.
- Proposed *Relative-Reward Matching*, learning from relative reward feedback while retaining a pretrained reference and preventing unnecessary updates at pretrained initialization when outputs receive equal rewards.

Secure Language-Model Agents [8]

- Developed *CoDA*, enforcing **trusted task policies throughout diffusion generation** to keep agents' tool calls and sensitive arguments within authorized workflows under prompt injection.
- Led static and adaptive prompt-injection evaluations across two diffusion language models and four AgentDojo domains; observed **zero successful adaptive attacks with correctly specified tool and argument permissions**, under budgets of up to 100 queries per scenario.

Selected Publications

Research output: 11 peer-reviewed papers published or accepted; 116 citations (Oct. 5, 2026). Selected works below. *Equal contribution.

[1] **Chenrui Ma**, Xi Xiao, Lin Zhao, Tianyang Wang, Ferdinando Fioretto, and Yanning Shen. (2026). *Drift Flow Matching*. **NeurIPS 2026**. [Paper] [Project] [Code]

[2] **Chenrui Ma**, Xi Xiao, Tianyang Wang, Xiao Wang, and Yanning Shen. (2026). *Stochastic Interpolants via Conditional Dependent Coupling* (CDC). **NeurIPS 2026**. [Paper]

[3] **Chenrui Ma**, Xi Xiao, Tianyang Wang, Xiao Wang, and Yanning Shen. (2026). *Learning Straight Flows: Variational Flow Matching for Efficient Generation* (S-VFM). **CVPR 2026**. [Paper]

[4] **Chenrui Ma**, Xi Xiao, Tianyang Wang, Xiao Wang, and Yanning Shen. (2026). *CAD-VAE: Leveraging Correlation-Aware Latents for Comprehensive Fair Disentanglement*. **AAAI 2026**. [Paper] [Code]

[5] Xi Xiao*, **Chenrui Ma***, Yunbei Zhang*, Chen Liu, Zhuxuanzi Wang, Yanshu Li, Lin Zhao, Guosheng Hu, Tianyang Wang, and Hao Xu. (2026). *Not All Directions Matter: Towards Structured and Task-Aware Low-Rank Model Adaptation*. **ACL 2026**. [Paper] [Code]

[6] Xinru Jiang*, Lin Zhao*, Xi Xiao, Yunbei Zhang, Janet Wang, **Chenrui Ma**, Haolin Li, Yanzhi Wang, Yifan Gong, and Octavia Camps. (2026). *Dynamic Hub-and-Spoke Memory for Streaming Video Understanding*. **Findings of EMNLP 2026**. [Paper]

[7] **Chenrui Ma**, Xi Xiao, Lin Zhao, Xinru Jiang, Tianyang Wang, Yanning Shen, and Ferdinando Fioretto. (2026). *Posterior Flow Matching*. **Preprint**. [Paper] [Project] [Code]

[8] Michael Cardei, Anjun Hu, **Chenrui Ma**, Francesco Pinto, Chawin Sitawarin, Chulin Xie, and Ferdinando Fioretto. (2026). *Only What You've Signed up For: Secure Workflows with Constrained Diffusion Agents* (CoDA). **Preprint**. [Project]

Research Experience

University of Virginia — RAISE Group *August 2026 – Present*
Graduate Research Assistant, advised by Prof. Ferdinando Fioretto; collaboration began May 2026.

University of California, Irvine — Trustworthy ML/AI Group *September 2024 – June 2026*
Research Assistant, advised by Prof. Yanning Shen.

External Research Collaboration — Oak Ridge National Laboratory *May 2025 – Present*
Research collaborator and coauthor with Dr. Xiao Wang in generative modeling and scientific ML.

Teaching: Teaching Assistant, University of California, Irvine January 2026 – March 2026

Technical Skills

Machine Learning: Diffusion/flow-based generative modeling; reinforcement learning; generative-model post-training.
Mathematical Methods: Probabilistic modeling; optimization; stochastic processes; ordinary and stochastic differential equations.
Programming and Frameworks: Python, PyTorch, JAX.
Systems: GPU computing, Slurm/HPC, Linux, Git, Docker.

Academic Service and Selected Honors

Co-organizer & Area Chair: Latent Visual Reasoning (LVR) Workshop, WACV 2027.
Reviewer: NeurIPS 2026 (**Top Reviewer**); ICLR 2027; AAAI 2026, 2027; *npj Digital Medicine* (2025).
Honors: UVA Computer Science Scholar (2026); AAAI-26 Student Scholarship and Volunteer Award (2025).